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Magnetometry analysis via a multicomponent Langevin-Weiss model with susceptibility-dependent demagnetization

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ABSTRACT

This study introduces a reformulated multicomponent Langevin-Weiss mean-field model for the analysis of magnetometry data. By expressing the model in terms of experimentally accessible quantities –mass magnetization and external magnetic field rather than volume magnetization and true magnetic field– the approach enables consistent analysis from low-field region to saturation. Finite element simulations are employed to account for susceptibility-dependent field effects arising from non-ellipsoidal sample geometries, and a compact parametrization of the resulting demagnetizing factors enables efficient correction without iterative numerical simulations. The methodology is effectively applied to a melt-spun Ni-Co-Mn-Sn Heusler alloy, capturing its complex magnetic behavior and resolving distinct ferromagnetic and paramagnetic contributions. The results further reveal an intrinsic effective magnetic anisotropy of the sample.

1. Introduction

Magnetometry, typically performed using vibrating sample magnetometers (VSMs) or superconducting quantum interference devices (SQUIDs) [1], is a key technique for characterizing the magnetic properties of materials ranging from ferro and ferrimagnets to paramagnets. Careful analysis of magnetometry data allows one to determine intrinsic parameters such as the saturation magnetization (M_S), initial susceptibility (χ_{init}), and effective anisotropy constant (K_{eff}). These quantities are essential both for advancing fundamental understanding and for guiding technological applications of magnetic materials, such as magnetic nanoparticle hyperthermia, magnetic particle imaging, magnetic refrigeration, and the development of sensors and actuators [2].

Despite its widespread use, extracting these parameters from magnetometry data is not straightforward and remains a source of inconsistency in the literature. A principal source of ambiguity arises because commonly used magnetization models to fit the data, such as the law of approach to saturation or Langevin-type functions, are formulated in terms of the volume magnetization M and the magnetic field H of the magnetized specimen. In contrast, magnetometers directly measure the total magnetic moment of the sample, from which the average mass (specific) magnetization σ [3,4] is typically derived, together with the externally applied magnetic field H_{ext}. Neglecting this distinction between measured and modeled quantities can lead to systematic errors, particularly when demagnetizing field effects are treated using geometry-dependent constants that neglect their susceptibility dependence.

In this work, we address these challenges by reformulating a multicomponent Langevin-Weiss mean-field model in terms of magnetometric quantities (σ and H_{ext}), enabling accurate fits of magnetometry

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data from low fields to saturation. The approach directly yields the saturation mass magnetization and initial apparent mass susceptibility, which can be converted to volume quantities when the mass density is known. We introduce a method to obtain susceptibility-dependent magnetometric demagnetizing factors without iterative finite-element calculations, allowing accurate recovery of the intrinsic susceptibility and effective anisotropy energy density. We illustrate the method using a Ni-Co-Mn-Sn Heusler alloy, a system of considerable interest for its multifunctional properties, including magnetic shape memory behavior, superelasticity, magnetocaloric effect, and magnetoelastic coupling [5–9].

2. The demagnetizing field

The purpose of this section is to clarify the role of the demagnetizing field in magnetometry analysis and to highlight the limitations of existing approximations when applied to materials with finite susceptibility.

In a finite magnetic body placed in a uniform external magnetic field $\boldsymbol{H}_{ext}$, the normal component of the magnetization $\boldsymbol{M}$ at the surface generates a magnetic field that tends to oppose $\boldsymbol{M}$ within the volume. Spatial variations of $\boldsymbol{M}$ within the body further contribute to this field. This internal magnetic field, known as the demagnetizing field, $\boldsymbol{H}_d$, is central to the interpretation of magnetization measurements.

For an ellipsoidal sample, both $\boldsymbol{M}$ and $\boldsymbol{H}_d$ are uniform throughout the body. This uniformity allows the definition of the demagnetizing factor, which arises naturally as the proportionality symmetric 3 × 3 tensor given by

$$\boldsymbol{H}_{\mathrm{d}} = -\boldsymbol{N}_d \boldsymbol{M}. \quad (1)$$

In its principal-axis frame, $\boldsymbol{N_d}$ is diagonal with components $N_{d,x}$, $N_{d,y}$, and $N_{d,z}$, which satisfy the sum rule [10].

$$N_{d,x} + N_{d,y} + N_{d,z} = 1. \quad (2)$$

The magnetic field within an ellipsoid is therefore the superposition of the externally applied field and the internal demagnetizing field, this is,

$$\boldsymbol{H} = \boldsymbol{H}_{ext} + \boldsymbol{H}_d = \boldsymbol{H}_{ext} - \boldsymbol{N}_d \boldsymbol{M}. \quad (3)$$

Differentiating Eq. (3) with respect to the magnetization along a principal axis yields a relationship between the isothermal intrinsic magnetic susceptibility $\chi = \partial M / \partial H$, the apparent (external) susceptibility $\chi_{app} = \partial M / \partial H_{ext}$, and the corresponding principal demagnetizing factor, namely

$$\chi^{-1} = \chi_{app}^{-1} - N_d. \quad (4)$$

For non-ellipsoidal geometries, however, the demagnetizing field is generally nonuniform, even under a uniform applied field. Consequently, for a material with finite, field-independent susceptibility, the magnetization also becomes nonuniform, and the spatial distribution of fields depends both on the body's geometry and χ. In this case, an effective magnetometric demagnetizing factor along the direction of the applied field (denoted u) can be defined as

$$N_{d,u} = -\langle H_{d,u} \rangle / \langle M_u \rangle, \quad (5)$$

where $\langle \bullet \rangle$ denotes volume averaging. The corresponding apparent susceptibility is

$$\chi_{app} = \langle M_u \rangle / H_{ext}. \quad (6)$$

Unlike in ellipsoids, the sum of the three orthogonal effective magnetometric demagnetizing factors in non-ellipsoidal samples may deviate from unity [11]. By numerical computation we found that $N_{d,x} + N_{d,y} + N_{d,z}$ decreases with increasing χ (Fig. 1), consistent with the loss of magnetization uniformity in non-ellipsoidal shapes (Fig. 2).

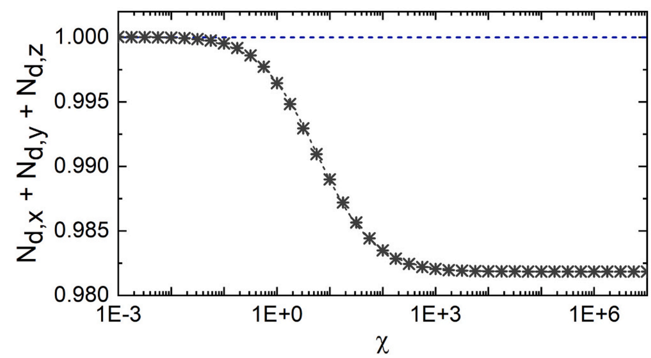

Fig. 1. Sum of the magnetometric demagnetizing factors along the three principal directions of a rectangular prism (2.72 mm × 2.08 mm × 20 μm), computed numerically by finite element analyses for a wide range of finite susceptibilities. The decrease of $N_{d,x} + N_{d,y} + N_{d,z}$ below unity with increasing χ reflects the progressively nonuniform magnetization that arises in non-ellipsoidal geometries.

Analytical expressions for $\langle H_{d,u} \rangle$ in some non-ellipsoidal geometries can be obtained in the limit $\chi \rightarrow 0$, where the magnetization is approximately uniform (Fig. 2). In this limit, evaluating Eq. (5) reduces to computing and volume-averaging the demagnetizing field generated by a uniformly magnetized body [12]. Closed-form solutions for the corresponding magnetometric demagnetizing factors are available for specific shapes such as cylinders [13] and rectangular prisms [14]. These results are widely used in magnetometry to estimate the internal "true" magnetic field from the applied field [3], often referred to as "the shearing correction" [15].

However, although the demagnetizing field effect has been studied for over a century, recent literature often omits a key limitation: these classical closed-form approximations rely on the assumption of an approximately uniform magnetization, which is only rigorously justified in the limit of infinitesimally small susceptibility for non-ellipsoidal shapes (e.g., [14]). Although the approximations are frequently applied to soft magnets far from saturation, this assumption is only strictly valid for typical paramagnets ($10^{-5} \lesssim \chi \lesssim 10^{-2}$), diamagnets ($\chi \approx -10^{-5}$), or for saturated ferromagnets, where the differential susceptibility is small. Paradoxically, classical approximations precisely fail in the high-susceptibility limit, where accurate demagnetization corrections are most important. By contrast, as $\chi \rightarrow 0$ ($\chi^{-1} \rightarrow \infty$), the term N_d in Eq. (4) becomes negligible compared to χ^{-1}, so χ approaches χ_{app} (Fig. 3) and the result becomes insensitive to the uncertainty in N_d. Therefore, for an accurate determination of the intrinsic susceptibility in non-ellipsoidal samples, numerical methods are generally required to compute effective magnetometric demagnetizing factors that account for both geometry and susceptibility.

Failure to recognize that the accuracy of classical closed-form approximations deteriorates away from the limit $\chi \rightarrow 0$ can introduce systematic errors. For instance, several studies have recently reported unphysical "overskewed" magnetization curves that exhibit negative initial susceptibilities after application of the standard shearing correction [16–19]. A related issue concerns the sample geometries selected in an effort to avoid variations in shape anisotropy across measurement directions, such as along the principal axes of the squared base in square prisms [20], or along different in-plane angles in disc-shaped samples [21]. Yet, for materials with in-plane magnetic anisotropy, the demagnetizing field itself becomes anisotropic: it depends not only on geometry but also on the susceptibility tensor, and therefore on the measurement direction within the plane.

Furthermore, since most magnetic measurements involve nonlinear magnetization curves, care is required when applying demagnetization corrections based on a constant χ. In a linear material, the spatial distributions of $\boldsymbol{M}$ and $\boldsymbol{H}$, and thus the magnetometric demagnetizing fac-

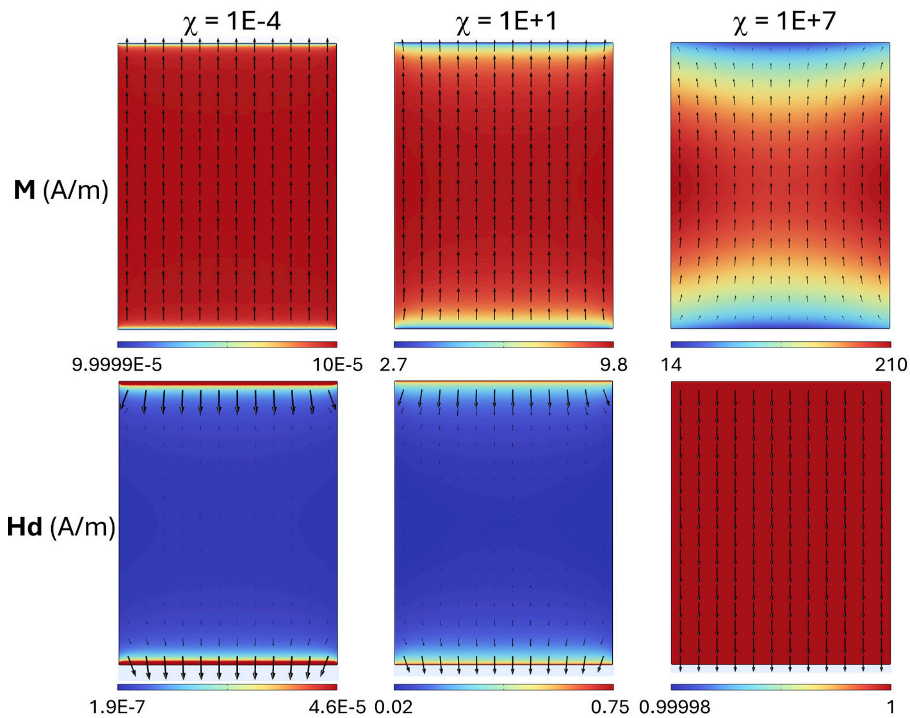

Fig. 2. Spatial distributions of magnetization $\boldsymbol{M}$ and demagnetizing field $\boldsymbol{H_d}$ in the midplane of a rectangular prism, computed by finite element analyses for low, intermediate, and high values of χ. Low χ results in an almost uniform $\boldsymbol{M}$, whereas high χ results in an almost uniform $\boldsymbol{H_d}$. An external field of 1 A/m is applied along the longitudinal axis (upwards). Prism dimensions: 2.72 mm × 2.08 mm × 20 μm.

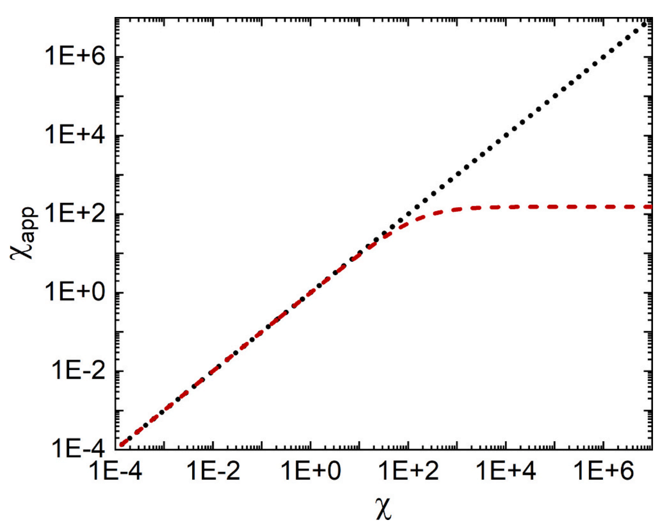

Fig. 3. Apparent susceptibility χ_{app} versus intrinsic susceptibility χ for a rectangular prism (2.72 mm × 2.08 mm × 20 μm) subjected to an external field applied along the longitudinal axis, obtained from 3D magnetostatic finite element simulations. In the limit $\chi \to 0$, the difference between χ_{app} and χ becomes negligible.

tor, are determined solely by the specimen geometry and intrinsic susceptibility (Fig. 2). They remain independent of H_{ext} because both the numerator and denominator in Eq. (5) scale linearly with the applied field. In contrast, in a nonlinear material, the functional form of $M(H)$ affects the internal field differently at various points within the body. This modifies the fields distributions and, consequently, the effective demagnetization factor, making iterative numerical procedures necessary for an accurate determination of the magnetization behavior [11,22]. In practice, however, demagnetizing field corrections are most critical at low fields, where initial susceptibilities are determined, and become relatively less important (although not necessarily small) as the sample approaches saturation [15]. Thus, first-order susceptibility-based corrections are often sufficient.

Accordingly, the determination of effective demagnetizing factors suitable for correcting magnetization measurements in non-ellipsoidal samples –under the simplifying assumption of a constant intrinsic susceptibility– has motivated extensive numerical work, comprehensively reviewed by Chen [11]. In particular, Chen and co-workers constructed susceptibility-dependent datasets of effective demagnetizing factors for experimentally relevant geometries, including cylinders [11,13] and rectangular prisms [23], spanning a broad range of aspect ratios. These studies further introduced an interpolation procedure that allows the evaluation of $N_d(\chi)$ for arbitrary intermediate aspect ratios within the explored parameter space.

Once a susceptibility-dependent dataset $N_d(\chi)$ is available for a given geometry, the application of a demagnetization correction introduces an intrinsic inverse problem, because the intrinsic susceptibility χ is not known a priori: only the apparent susceptibility χ_{app} can be directly extracted from the measured magnetization curve. Within the linear, constant-χ framework, χ and N_d must therefore satisfy an implicit self-consistency relation, obtained by rearranging Eq. (4), $\chi = \left(\chi_{app}^{-1} - N_d(\chi)\right)^{-1}$. Chen and co-workers addressed this problem by introducing a fixed-point iteration scheme: an initial estimate $\chi^{(0)}$ (typically based on the $\chi \to 0$ limit) is used to evaluate $N_d\left(\chi^{(0)}\right)$ from the tabulated or interpolated datasets, followed by successive updates $\chi^{(i+1)} = \left(\chi_{app}^{-1} - N_d\left(\chi^{(i)}\right)\right)^{-1}$, until convergence is reached. Using this procedure, effective demagnetizing factors were reported with relative accuracies of up to ~0.3% [11,22,23].

However, as recently demonstrated by Etse and Mininger [18], even demagnetizing factors determined with such high accuracy can still lead to substantial errors in the inferred initial intrinsic susceptibility when applied to soft magnetic materials. As discussed above, uncertainty in N_d

becomes increasingly critical as χ grows, amplifying small residual errors in the demagnetizing correction. To mitigate this limitation, Ref. [18] proposed a fully numerical self-consistent approach based on iterative recalculation of the magnetostatic field, allowing the susceptibility error to be controlled within a prescribed tolerance. Using this method, the authors demonstrated excellent agreement between an open-circuit magnetization curve corrected with a susceptibility-dependent demagnetizing factor and a closed-circuit measurement over the entire field range, even in the presence of strongly nonlinear responses (see Fig. 3(b) of Ref. [18]).

The approach adopted in this work follows the same linear-susceptibility demagnetization framework –which enables seamless integration with the multicomponent Langevin-Weiss mean-field model– but differs in its implementation and computational cost. Instead of relying on iterative lookups in dense numerical tables during each inversion step, or on iterative solutions of the full magnetostatic finite element model, we compute $N_d(\chi^{-1})$ once for the specific specimen geometry using finite-element simulations over a broad range of χ^{-1}. This dependence is then captured by a compact analytical parameterization. The resulting surrogate model reduces the inverse demagnetization determination to a simple scalar root-finding problem, which can be solved both rapidly and accurately using standard numerical methods. This strategy yields a practical and robust workflow for demagnetization correction and intrinsic susceptibility extraction in magnetometry of non-ellipsoidal samples.

3. Isothermal magnetization models

Several models are employed to interpret isothermal magnetization data. The law of approach to saturation,

$$M(H) = M_S\left(1 - \frac{a'}{H} - \frac{b'}{H^2}\right) + \chi' H, \tag{7}$$

is widely used to fit the high-field region, allowing estimates of M_S [2,24–30] and, under appropriate assumptions, estimates of K_{eff} from the $1/H^2$ term [2,8,24,26,31]. However, its applicability is restricted to fields near saturation, a region for which there is no standardized method of determination. Alternatively, modified Langevin models, e.g. [26,32–36],

$$M(H) = M_S\,\mathscr{L}\left(\frac{H}{a}\right) + \chi' H, \tag{8}$$

where $\mathscr{L}(h) = \coth(h) - 1/h$ is the Langevin function [37], extend the fitting to lower fields by describing the system as an ensemble of noninteracting magnetic clusters, as in superparamagnets. These models often fail, however, in real materials where interactions are non-negligible [26,32].

A multicomponent Langevin-Weiss mean-field model has recently been proposed to describe systems comprising both noninteracting and interacting effective magnetic moments, offering a better description of anhysteretic magnetization curves for a broad range of materials, from the low-field Rayleigh region to near-saturation [38]. In this model, the system is described as a mixture of components, each contributing independently to the total magnetization:

$$M(H) = \sum_i^n M_i(H) = \sum_i^n M_{S,i}\,\mathscr{L}\left(\frac{H_{eff,i}(H)}{a_i}\right), \tag{9}$$

where $M_{S,i}$ is the saturation magnetization of the i-th component, $\mathscr{L}$ is the Langevin function, and $a_i = N_i k_B T/(\mu_0 M_{S,i})$ is a temperature-dependent scaling parameter (N_i is the number density of elementary magnetic moments in thermal equilibrium at temperature T, k_B is the Boltzmann constant, and μ_0 is the vacuum permeability) [39]. As in Weiss' single-component description, this model treats the magnetic medium as effectively homogeneous without considering the spatial distribution of magnetic moments [39]. By contrast, for dilute magnetic particle systems or strongly heterogeneous multiphase materials, a mesoscopic treatment of spatial distributions of particles/phases may be necessary, since magnetodipolar interactions produce spatially varying internal fields that depend sensitively on interparticle distances, granularity, and spatial clustering [40–42].

A component with very large a_i describes an almost linear term, analogous to the $\chi' H$ term in Eqs. (7) and (8), and can represent a paramagnetic phase or forced magnetization [1]. The effective field acting on each component includes a first-order Weiss-type internal contribution,

$$H_{eff,i} = H + \alpha_i M_i. \tag{10}$$

where α_i quantifies an effective self-interaction strength that can be either positive or negative [43]. Although it is possible to expand Weiss' field into higher odd powers of M_i [44], retaining only the first-order term maintains simplicity and has proven sufficient for accurate descriptions of magnetization data.

In contrast to the law of approach to saturation or simple Langevin-based models, the Langevin-Weiss mean-field approach can naturally incorporate a first-order approximation of the demagnetizing field present in magnetometric measurements:

$$H_{eff,i} = H_{ext} - N_{d,i} M_i + \alpha_i M_i. \tag{11}$$

where $N_{d,i}$ depends on both the sample geometry and the susceptibility of the i-th component –specifically, its initial susceptibility $\chi_{init,i}$ in this linear approximation.

Finally, assuming a uniform and constant mass density ρ throughout the measurement range, the multicomponent Langevin-Weiss model in magnetometric terms is given by

$$\sigma(H_{ext}) = \sum_i^n \sigma_i(H_{ext}) = \sum_i^n \sigma_{S,i}\,\mathscr{L}\left(\frac{H_{ext} + \beta_i \sigma_i}{a_i}\right). \tag{12}$$

where $\beta_i = (\alpha_i - N_{d,i})\rho$ and σ_i is the mass magnetization of the i-th component.

This formulation [Eq. (12)] enables the direct fitting of the sample's average mass magnetization as a function of the external magnetic field, allowing the estimation of the total saturation mass magnetization as

$$\sigma_S = \sum_i^n \sigma_{S,i}, \tag{13}$$

and the initial apparent mass susceptibility as (see analogue derivation for χ_{init} in [43])

$$\chi_{m,init}^{app} = \left.\frac{\partial \sigma_i}{\partial H_{ext}}\right|_{H_{ext}\to 0} = \sum_i^n \frac{\sigma_{S,i}}{3a_i - \beta_i \sigma_{S,i}}. \tag{14}$$

Furthermore, the effective anisotropy energy per unit volume is often estimated from the graphical method of computing the area of the triangle delimited by the vertical line at $H = 0$ and the tangents to the low- and high-field regions of the $M(H)$ curve [15], i.e.,

$$E = \frac{1}{2}\mu_0 M_S^2 \chi_{init}^{-1}, \tag{15}$$

and can be written in terms of the model parameters noticing that the saturation magnetization is

$$M_S = \rho \sigma_S \tag{16}$$

and that the initial intrinsic susceptibility inverse is given by

$$\chi_{init}^{-1} = \left(\sum_{i}^{n} \left(\frac{3a_i}{\rho \sigma_{S,i}} - \alpha_i \right)^{-1} \right)^{-1}. \tag{17}$$

Eq. (17) follows from Eq. (14) by considering that $\chi_{init} = \rho \chi_{m,init}$ and that $\chi_{m,init} = \sum_i^n \frac{\sigma_{S,i}}{3a_i - \alpha_i \rho \sigma_{S,i}}$.

4. Methods

A Heusler alloy with nominal composition $Ni_{42}Co_8Mn_{39}Sn_{11}$ (at. %) was synthesized as a polycrystalline ribbon via melt-spinning. A segment of the ribbon, measuring 2.72 mm in length, 2.08 mm in width, and 20 μm in thickness, was used for magnetic characterization. Measurements were conducted at 295 K using a vibrating sample magnetometer (VSM-VersaLab, Quantum Design) under applied magnetic fields of up to 2387 kA/m (30 kOe).

Two measurement configurations were employed: (i) in-plane, along the ribbon casting direction (hereafter referred to as "longitudinal"), and (ii) out-of-plane, normal to the ribbon surface (hereafter referred to as "normal"). Each experimental mass magnetization curve was analyzed using the MagAnalyst toolbox [45], which applies the multicomponent Langevin-Weiss model to fit anhysteretic magnetization curves. In practice, anhysteretic curves can be approximated by averaging the left and right branches of symmetric hysteresis loops. When fitting $\sigma(H_{ext})$ curves, rather than $M(H)$, the parameters extracted with the toolbox correspond to $\sigma_{S,i}$, a_i, and β_i, rather than $M_{S,i}$, a_i, and α_i. The theoretical mass density of the alloy, estimated using the rule of mixtures, was $\rho \approx 7.91$ g/cm^3. Magnetometric demagnetizing factors for both configurations were estimated through finite-element magnetostatic simulations, as described in the next section.

5. Results and discussion

The mass magnetization curves of $Ni_{42}Co_8Mn_{39}Sn_{11}$, measured along both the longitudinal and normal directions at 295 K, exhibit negligible hysteresis (Fig. 4). The fitted anhysteretic curves, presented in both standard format (Fig. 5) and semi-logarithmic derivative format (Fig. 6), show excellent agreement with the experimental data across the entire applied field range. The retrieved parameters, along with their relative fitting errors, are listed in Table I. Within the applied field range, only the linear region of component 0 is captured. Consequently, the fit is underdetermined for this component, and one of the three model parameters was held fixed in the toolbox to fit the remaining two. Accordingly, for component 0 we report $\chi_{m,init,0}^{app}$ [computed from the corresponding term in Eq. (14)] rather than the model parameters themselves.

The semi-logarithmic derivative representation (Fig. 6) reveals subtle inhomogeneities in the magnetic response, evidenced by overlapping contributions from multiple components. In this representation, a single Langevin component would yield a well-defined halo [39], which is insufficient to capture the complexity observed in the experimental data. The dominant contribution (component 1) is attributed to the ferromagnetic austenitic phase, whose high crystallographic symmetry favors stronger coupling between magnetic moments than the lower-symmetry martensitic phase [46]. The martensitic phase, which exhibits weaker ferromagnetism [9], is likely responsible for the second contribution (component 2). A small martensitic fraction in this system near room temperature has already been reported in related studies [8,32,47]. A third contribution (component 0), characterized by an approximately linear response over the analyzed field range, accounts for the nearly linear increase in the $\partial\sigma/\partial(lnH_{ext})$ versus H_{ext} curves at high fields in the log-log representation. This behavior most likely corresponds to a residual paramagnetic phase that remains stable at 295 K and does not saturate within the applied field range.

Therefore, the total saturation mass magnetization, was obtained by summing the contributions of components 1 and 2, and was estimated to be $\sigma_S = 107 \pm 1$ Am^2/kg. This value was consistent across both measurement directions, indicating a macroscopically isotropic saturation behavior [1,48]. However, the applied magnetic field required to reach saturation was negligibly small in the longitudinal direction and significantly larger in the normal direction. This anisotropy can be understood by considering the sample's geometry, which favors in-plane magnetization. To analyze the intrinsic behavior of the material, we carried out a three-dimensional finite element analysis (FEA) that accounts for the effects of both geometry and susceptibility on the demagnetizing field.

The FEA model consisted of a rectangular prism with the same dimensions as the physical sample measured in the VSM, surrounded by a large air region. Filleted edges were incorporated, and a convergence study was performed to ensure mesh stability. The sample was assigned a constant scalar susceptibility χ and subjected to a uniform external field ($H_{ext} = 1$ A/m) applied along either the longitudinal or normal direction. Magnetostatic field computations were performed for χ values ranging from 10^{-4} to 10^7. The magnetometric demagnetizing factors were calculated using Eq. (5), and the apparent susceptibilities were

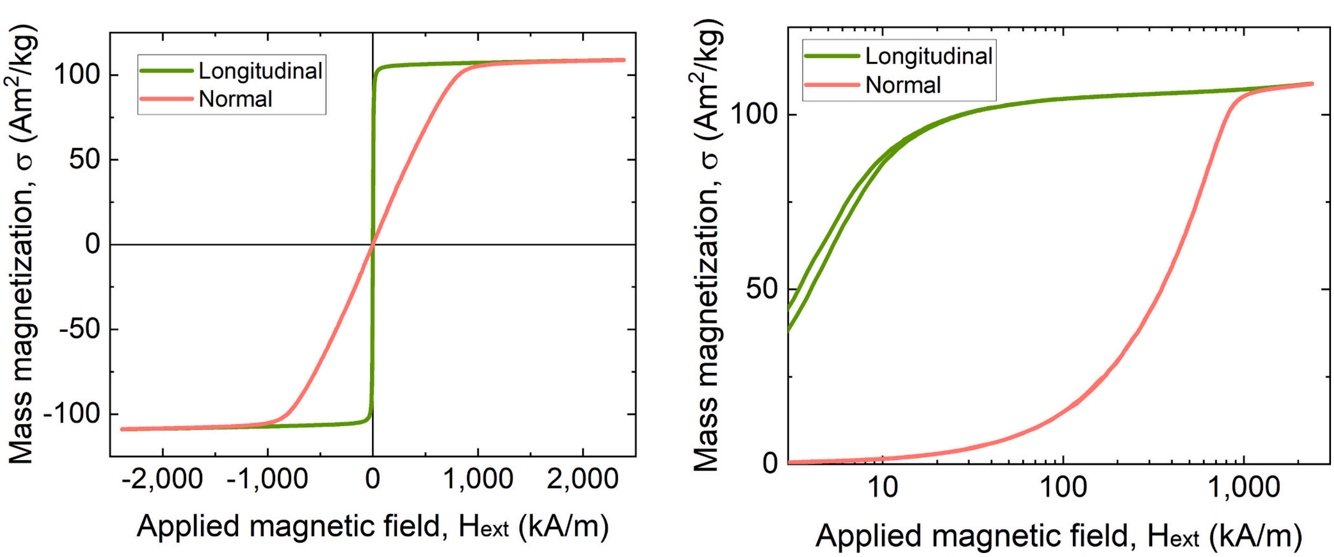

Fig. 4. Mass magnetization of $Ni_{42}Co_8Mn_{39}Sn_{11}$ at 295 K, measured in longitudinal and normal directions. Left: standard representation. Right: semi-logarithmic representation revealing small hysteresis in the longitudinal measurement.

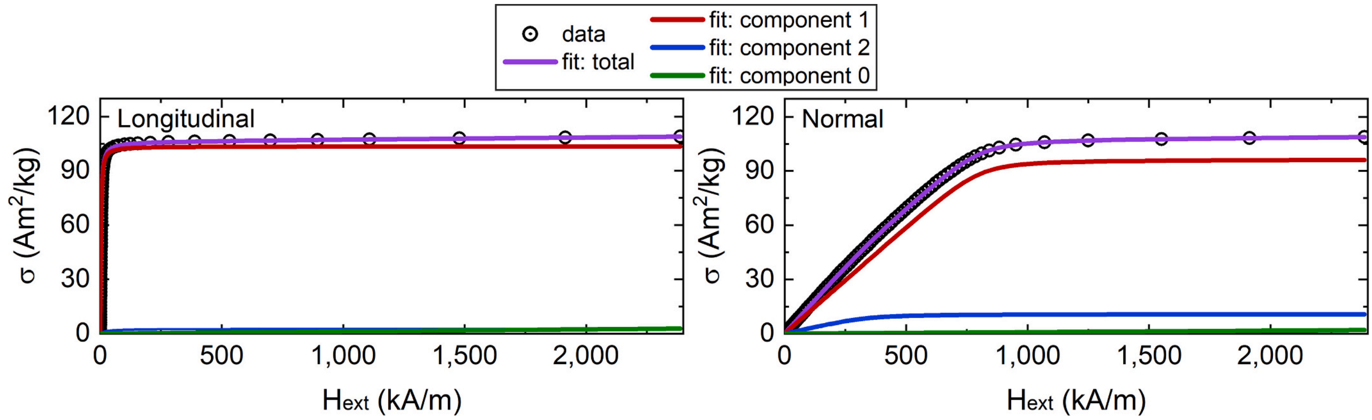

Fig. 5. Experimental anhysteretic mass magnetization curves and the corresponding fits using the multicomponent Langevin-Weiss model.

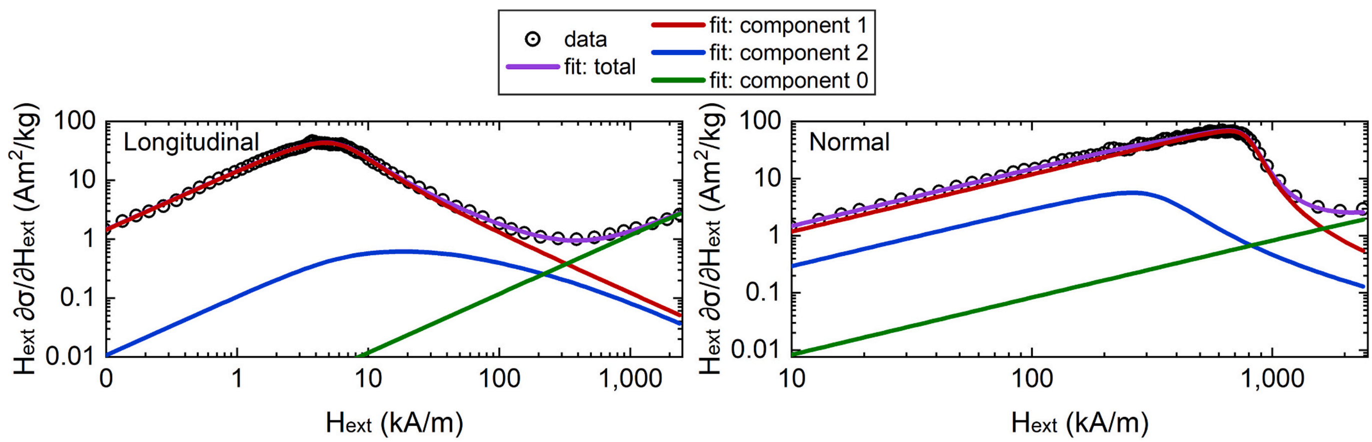

Fig. 6. Semi-logarithmic derivative of the anhysteretic mass magnetization, $(\partial\sigma/\partial(lnH_{ext}) = H_{ext}\ \partial\sigma/\partial H_{ext})$, and corresponding fitted curves.

Table I
Parameters of the multicomponent Langevin-Weiss model [Eq. (12)] obtained for the longitudinal and normal measurement directions: $\sigma_{S,i}$, β_i, and a_i, for components 1 and 2, and the initial apparent mass susceptibility for component 0. Relative errors correspond to the diagonal distance error as defined in Ref. [39].

Direction	$\sigma_{S,1}$ (Am^2/kg)	β_1 (kg/m^3)	a_1 (A/m)	$\sigma_{S,2}$ (Am^2/kg)	β_2 (kg/m^3)	a_2 (A/m)	$\chi^{app}_{m,init,0}$(m^3/kg)	Relative error
Long.	103.547	−35	1175	2.692	29,529	34,942	1.17×10^{-6}	4.39×10^{-5}
Normal	96.466	−8271	6028	10.873	−27,938	21,740	8.31×10^{-7}	2.74×10^{-5}

determined via Eq. (6). Fig. 7 shows the dependence of N_d and χ_{app}^{-1} on χ^{-1} for the longitudinal and normal configurations, along with Aharoni's classical approximations for N_d [14].

We found that the susceptibility-dependent magnetometric demagnetizing factor obtained from the FEA is well represented by the logistic form,

$$N_d = N_{d,max} - \frac{N_{d,max} - N_{d,min}}{1 + \left(\chi^{-1}/\chi_0^{-1}\right)^p}, \tag{18}$$

where the fitted parameters $N_{d,min}$, $N_{d,max}$, χ_0^{-1}, and p are reported in Table II. The fits yield excellent coefficients of determination for both fits: $R^2 > 0.9995$. The high R^2 values show that the χ-sweep FEA may be conducted on a substantially sparser χ grid, reducing computational effort without compromising fit accuracy. This parametrization captures the physically expected asymptotes: $N_d \rightarrow N_{d,min}$ for $\chi^{-1} \rightarrow 0$ $(\chi \rightarrow \infty)$ and $N_d \rightarrow N_{d,max}$ for $\chi^{-1} \rightarrow \infty$ $(\chi \rightarrow 0)$. When plotted on a logarithmic scale of χ^{-1}, Eq. (18) appears as a sigmoid with its midpoint set by χ_0^{-1} and its steepness controlled by p. Eq. (18) can be analytically inverted to express χ^{-1} as a function of N_d,

$$\chi^{-1} = \chi_0^{-1}\left(\frac{N_d - N_{d,min}}{N_{d,max} - N_d}\right)^{\frac{1}{p}}. \tag{19}$$

Combining Eq. (19) with the internal field relation [Eq. (4)] yields the implicit connection between the magnetometric demagnetizing factor and the apparent susceptibility,

$$\begin{aligned} \chi_{app}^{-1} &= \chi^{-1} + N_d, \\ \chi_{app}^{-1} &= \chi_0^{-1}\left(\frac{N_d - N_{d,min}}{N_{d,max} - N_d}\right)^{\frac{1}{p}} + N_d. \end{aligned} \tag{20}$$

Because Eq. (20) cannot be solved in closed form for N_d, we determine N_d numerically for a known apparent susceptibility inverse χ_{app}^{-1}. In practice, χ_{app}^{-1} is obtained from the mass density and the inverse low-field slope of the anhysteretic response [computed from Eq. (14) using the multicomponent Langevin-Weiss model parameters],

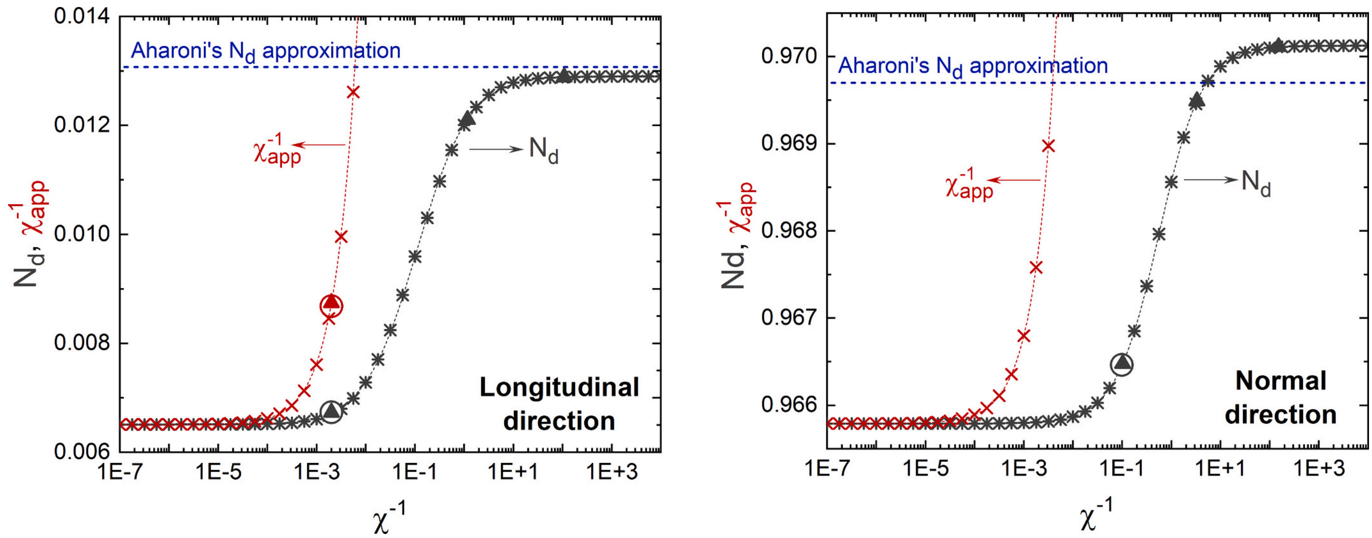

Fig. 7. Magnetometric demagnetizing factor N_d and apparent susceptibility inverse χ_{app}^{-1} as functions of the intrinsic susceptibility inverse χ^{-1} for a rectangular prism (2.72 mm × 2.08 mm × 20 μm), computed using a 3D magnetostatic finite element model. Results are shown for the longitudinal (left) and normal (right) measurement directions. A logistic fit describes $N_d(\chi^{-1})$ [Eq. (18)]. Solid triangles mark the solutions corresponding to model components 1, 2, and 0; open circles denote the total response. The dashed horizontal line indicates Aharoni's analytical estimate of N_d for this geometry [14].

Table II
Fitted parameters of the logistic model for the magnetometric demagnetizing factor $N_d(\chi^{-1})$ [Eq. (18)] for the longitudinal and normal measurement directions.

Direction	$N_{d,min}$	$N_{d,max}$	χ_0^{-1}	p
Long.	0.0065	0.0129	0.1096	0.8185
Normal	0.9658	0.9701	0.5595	0.9856

$\chi_{app}^{-1} = \left(\rho\chi_{\substack{app\\ m,init}}\right)^{-1}$. For each direction, we solve Eq. (20) for N_d using a scalar root-finding routine (e.g., MATLAB's `fzero`) constrained to the physically admissible interval $N_{d,min} + \varepsilon < N_d < N_{d,max} - \varepsilon$. Once N_d is obtained, the intrinsic susceptibility follows directly from Eq. (4) (equivalently from Eq. (19)). The resulting N_d values and intrinsic susceptibilities (computed both for the total response and for each fitted component) are marked in Fig. 7 for both measurement directions.

Accounting for the coupled dependence of N_d on both geometry and susceptibility was essential for the longitudinal configuration. In this case, using Aharoni's N_d approximation, which overestimated N_d by 94% for our sample, yielded an unphysical negative value of χ_{init}. In contrast, the normal direction was comparatively insensitive to this refinement, as the difference between the classical and corrected estimates of N_d and χ_{init} was small.

Even after accounting for demagnetizing effects, the sample exhibits an easier in-plane magnetization. The corresponding effective anisotropy energy densities are $(4.5 \pm 0.3) \times 10^4$ J/m^3 for the normal direction and $(9.0 \pm 0.6) \times 10^2$ J/m^3 for the longitudinal direction. This outcome is not trivially anticipated: the dominant austenitic phase is cubic and therefore does not favor either direction, and the high cooling rate during melt spinning typically induces a fine-grained texture elongated in the normal direction [26,49]. Nevertheless, our results indicate that the alloy is more easily magnetized along the ribbon's longitudinal direction.

6. Conclusions

We have reformulated the multicomponent Langevin-Weiss mean-field model for magnetometry analysis of effectively homogeneous magnetic systems. By fitting the full magnetometric response from low fields to saturation, the method enables reliable extraction of key magnetic parameters, including mass saturation magnetization and initial apparent mass susceptibility, with volume quantities obtained when the mass density is known.

Finite-element calculations were used to quantify the susceptibility dependence of magnetometric demagnetizing factors in non-ellipsoidal geometries. A compact parameterization of $N_d(\chi^{-1})$ allows intrinsic susceptibility to be obtained from apparent susceptibility without iterative finite-element simulations, reducing the demagnetizing factor determination to a simple scalar root-finding problem while preserving accuracy. This enables consistent estimation of anisotropy energy densities for different measurement orientations.

Classical analytical demagnetization corrections remain computationally simple but are valid only in the limit of nearly uniform magnetization and may yield unphysical results at high susceptibility. In this regime, the susceptibility-dependent correction scheme adopted here provides a robust and practical alternative. Application to a melt-spun Ni-Co-Mn-Sn Heusler alloy reveals a clear intrinsic effective magnetic anisotropy between longitudinal and normal ribbon directions, persisting after demagnetizing field corrections.

CRediT authorship contribution statement

Josefina M. Silveyra: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Andrés Rosales-Rivera:** Writing – review & editing, Validation, Resources, Project administration, Investigation, Funding acquisition. **Nicolás A. Salazar-Henao:** Investigation, Data curation. **Daniel Salazar:** Writing – review & editing, Validation, Resources, Investigation, Funding acquisition. **Juan M. Conde Garrido:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] S.A. Cavill, "Magnetic measurement instruments and techniques," IOP Magnetism Winter School. https://magnetism.eu/esm/2023/slides/esm2023-cavill-slides.pdf;, 2025. https://magnetism.eu/353-recordings.htm.
[2] K.S. Sivaranjani, G.A. Jacob, R.J. Joseyphus, Comprehensive law-of-approach-to-saturation for the determination of magnetic anisotropy in soft magnetic materials, Phys. Status Solidi B 259 (2022) 2200050, https://doi.org/10.1002/pssb.202200050.
[3] J.M.D. Coey, Magnetism and magnetic materials, Cambridge University Press, 2010, https://doi.org/10.1080/00107514.2010.514061.
[4] NIST (National Institute of Standards and Technology), "Units for magnetic properties, 2026," https://www.nist.gov/system/files/documents/pml/electromagnetics/magnetics/magnetic_units.pdf 01-FEB-2026.
[5] C.O. Aguilar-Ortiz, D. Soto-Parra, P. Álvarez-Alonso, P. Lázpita, D. Salazar, P. O. Castillo-Villa, et al., Influence of Fe doping and magnetic field on martensitic transition in Ni–Mn–Sn melt-spun ribbons, Acta Mater. 107 (2016) 9–16, https://doi.org/10.1016/j.actamat.2016.01.041.
[6] B. Rodríguez-Crespo, D. Salazar, S. Lanceros-Méndez, V. Chernenko, Development and magnetocaloric properties of Ni(Co)-Mn-Sn printing ink, J. Alloys Compd. 917 (2022) 165521, https://doi.org/10.1016/j.jallcom.2022.165521.
[7] M.N. Inallu, P. Kameli, A.G. Varzaneh, I.A. Sarsari, D. Salazar, I. Orue, et al., Magnetocaloric effect in W-doped Ni–Mn–Sn alloy probed by direct and indirect measurements, J. Phys. D. Appl. Phys. 52 (2019) 235001, https://doi.org/10.1088/1361-6463/ab0f79.
[8] L. Yi, T. Zhang, B. Zhang, Y. Xu, H. Piao, L. Pan, Induced magnetic anisotropy of Ni42Co8Mn39Sn11 Heusler alloy under magnetic-field-annealing, J. Magn. Magn. Mater. 495 (2020) 165843, https://doi.org/10.1016/j.jmmm.2019.165843.
[9] V. Srivastava, X. Chen, R.D. James, Hysteresis and unusual magnetic properties in the singular Heusler alloy Ni45Co5Mn40Sn10, Appl. Phys. Lett. 97 (2010), https://doi.org/10.1063/1.3456562.
[10] J.A. Osborn, Demagnetizing factors of the general ellipsoid, Phys. Rev. 67 (1945) 351, https://doi.org/10.1103/PhysRev.67.351.
[11] D.-X. Chen, J.A. Brug, R.B. Goldfarb, Demagnetizing factors for cylinders, IEEE Trans. Magn. 27 (1991) 3601–3619, https://doi.org/10.1109/20.102932.
[12] W.F. Brown, Magnetostatic principles in ferromagnetism, North-Holland Publishing Company, 1962.
[13] D.-X. Chen, E. Pardo, A. Sanchez, Fluxmetric and magnetometric demagnetizing factors for cylinders, J. Magn. Magn. Mater. 306 (2006) 135–146, https://doi.org/10.1016/j.jmmm.2006.02.235.
[14] A. Aharoni, Demagnetizing factors for rectangular ferromagnetic prisms, J. Appl. Phys. 83 (1998) 3432–3434, https://doi.org/10.1063/1.367113.
[15] B.D. Cullity, C.D. Graham, Introduction to magnetic materials, John Wiley & Sons, 2009, https://doi.org/10.1002/9780470386323.
[16] A. Panda, S. Basu, A. Mitra, Demagnetisation effect and its correction on the measurement of magnetic hysteresis loop of melt-spun ribbons, J. Magn. Magn. Mater. 261 (2003) 190–195, https://doi.org/10.1016/S0304-8853(02)01472-5.
[17] R. Skomski, Y. Liu, J.E. Shield, G.C. Hadjipanayis, D.J. Sellmyer, Permanent magnetism of dense-packed nanostructures, J. Appl. Phys. 107 (2010), https://doi.org/10.1063/1.3337657.
[18] K. Etse, X. Mininger, Determination of demagnetizing factors for various geometries using an iterative numerical approach, J. Magn. Magn. Mater. 564 (2022) 170151, https://doi.org/10.1016/j.jmmm.2022.170151.
[19] M. Zambach, M. Varón, M.R. Almind, M. Knaapila, Z. Ouyang, M. Beleggia, et al., Demagnetisation effects in single-domain particles, arXiv preprint (2025), https://doi.org/10.48550/arXiv.2502.03394.
[20] K. Karube, L. Peng, J. Masell, M. Hemmida, H.A. Krug Von Nidda, I. Kézsmárki, et al., Doping control of magnetic anisotropy for stable antiskyrmion formation in Schreibersite (Fe, Ni) 3P with S4 symmetry, Adv. Mater. 34 (2022) 2108770, https://doi.org/10.1002/adma.202108770.
[21] A. Srivastava, K. Cole, A. Wadsworth, T. Burton, C. Mewes, T. Mewes, et al., Broadband characterization of stress induced anisotropy in nanocomposite Co74.6Fe2.7Mn2.7Nb4Si2B14, J. Magn. Magn. Mater. 500 (2020) 166307, https://doi.org/10.1016/j.jmmm.2019.166307.
[22] D.-X. Chen, E. Pardo, Y.-H. Zhu, L.-X. Xiang, J.-Q. Ding, Demagnetizing correction in fluxmetric measurements of magnetization curves and hysteresis loops of ferromagnetic cylinders, J. Magn. Magn. Mater. 449 (2018) 447–454, https://doi.org/10.1016/j.jmmm.2017.10.069.
[23] D.-X. Chen, E. Pardo, A. Sanchez, Demagnetizing factors for rectangular prisms, IEEE Trans. Magn. 41 (2005) 2077–2088, https://doi.org/10.1109/TMAG.2005.847634.
[24] E.C. Devi, I. Soibam, Law of approach to saturation in Mn–Zn ferrite nanoparticles, J. Supercond. Nov. Magn. 32 (2018) 1293–1298, https://doi.org/10.1007/s10948-018-4823-4.
[25] V. Sánchez-Alarcos, J. López-García, I. Unzueta, J. Pérez-Landazábal, V. Recarte, J. Beato-López, et al., Magnetocaloric effect enhancement driven by intrinsic defects in a Ni45Co5Mn35Sn15 alloy, J. Alloys Compd. 774 (2019) 586–592, https://doi.org/10.1016/j.jallcom.2018.10.016.
[26] P. Czaja, J. Przewoźnik, M. Fitta, Heat treatment effect on the evolution of magnetic properties of martensite in magnetic shape memory Ni48Mn39.5Sn9.5Al3 Heusler alloy ribbons, Mater. Res. Bull. 135 (2021) 111120, https://doi.org/10.1016/j.materresbull.2020.111120.
[27] A. Abdallah, A. Aridi, M. Rabaa, R.M. Moussa, R. Awad, Tailored physical and magnetic properties by La3+ dopants in Mg–Ni–Co nanoferrites: insight into the law of approach to saturation, Appl. Phys. A Mater. Sci. Process. 129 (2023) 770, https://doi.org/10.1007/s00339-023-07040-y.
[28] N.G. Orsetti, M.A. Frechero, M.A.A. Manso, M.L. Montes, G. Suarez, M. Gamba, Aging of lithium zirconate (Li2ZrO3) at room temperature: effect of iron as a dopant, Ceram. Int. 51 (2025) 7470–7484, https://doi.org/10.1016/j.ceramint.2024.12.183.
[29] M.L.M. Elizalde, C. Acha, M.S. Moreno, P.S. Antonel, Tuning the electrical and magnetic properties in multifunctional composite materials based on the PEDOT: DBS conducting polymer and magnetite nanoparticles, J. Mater. Chem. C 10 (2022) 18264–18278, https://doi.org/10.1039/D2TC03175B.
[30] R. Ferrero, G. Barrera, F. Celegato, M. Vicentini, H. Sözeri, N. Yıldız, et al., Experimental and modelling analysis of the hyperthermia properties of iron oxide nanocubes, Nanomaterials 11 (2021) 2179, https://doi.org/10.3390/nano11092179.
[31] S. Andreev, M. Bartashevich, V. Pushkarsky, V. Maltsev, L. Pamyatnykh, E. Tarasov, et al., Law of approach to saturation in highly anisotropic ferromagnets application to Nd-Fe-B melt-spun ribbons, J. Alloys Compd. 260 (1997) 196–200, https://doi.org/10.1016/S0925-8388(97)00127-8.
[32] D. Cong, S. Roth, L. Schultz, Magnetic properties and structural transformations in Ni–Co–Mn–Sn multifunctional alloys, Acta Mater. 60 (2012) 5335–5351, https://doi.org/10.1016/j.actamat.2012.06.034.
[33] K. Cao, F. Tian, S. Huang, Y. Zhang, Q. Zhao, K. Yao, et al., Tuning the exchange bias effect via thermal treatment temperature in bulk Ni50Mn42In3Sb5 Heusler alloys, Appl. Phys. Express 14 (2021) 105502, https://doi.org/10.35848/1882-0786/ac225f.
[34] J. Guo, F. Tian, Q. Zhao, S. Kong, L. Xian, W. Zuo, et al., Different exchange bias behavior caused by tuning glassy state in Ni50Mn37Ga13 ribbons, J. Alloys Compd. 1002 (2024) 175238, https://doi.org/10.1016/j.jallcom.2024.175238.
[35] S. Salehizadeh, B. Costa, P. Sanguino, V. Rodrigues, J. Greneche, A. Cavaleiro, et al., Quantitative determination of surface spins contribution of magnetization, anisotropy constant, and cation distribution of manganese ferrite-silica nanocomposite, Mater. Sci. Eng. B Solid-State Mater. Adv. Technol. 284 (2022) 115902, https://doi.org/10.1016/j.mseb.2022.115902.
[36] S. Das, A. Banerjee, N. Bhakta, P. Chakrabarti, Preparation and investigations of the magnetoelectric properties of (Ni0.5Zn0.5Fe2O4)x [(Na0.5Bi0.5) 0.7Ba0.3TiO3]1−x (x= 0.3, 0.5) nanocomposite ceramics for magnetic field sensor applications, ACS Appl. Nano Mater. 7 (2024) 205–217, https://doi.org/10.1021/acsanm.3c04303.
[37] P. Langevin, Sur la théorie du magnétisme, J. Phys. Theor. Appl. 4 (1905) 678–693 (P. Langevin, Sur la théorie du magnétisme).
[38] J.M. Silveyra, J.M. Conde Garrido, A physically based model for soft magnets' anhysteretic curve, JOM 75 (2023) 1810–1823, https://doi.org/10.1007/s11837-023-05704-x.
[39] J.M. Silveyra, J.M. Conde Garrido, On the anhysteretic magnetization of soft magnetic materials, AIP Adv. 12 (2022), https://doi.org/10.1063/9.0000328.
[40] D. Berkov, N. Gorn, R. Schmitz, D. Stock, Langevin dynamic simulations of fast remagnetization processes in ferrofluids with internalmagnetic degrees of freedom, J. Phys. Condens. Matter 18 (2006) S2595, https://doi.org/10.1088/0953-8984/18/38/S05.
[41] P. Kharitonskii, N. Bobrov, K. Gareev, A. Kosterov, A. Nikitin, A. Ralin, et al., Magnetic granulometry, frequency-dependent susceptibility and magnetic states of particles of magnetite ore from the Kovdor deposit, J. Magn. Magn. Mater. 553 (2022) 169279, https://doi.org/10.1016/j.jmmm.2022.169279.
[42] P. Kharitonskii, E. Sergienko, A. Ralin, E. Setrov, T. Sheidaev, K. Gareev, et al., Superparamagnetism of artificial glasses based on rocks: experimental data and theoretical modeling, Magnetochemistry 9 (2023) 220, https://doi.org/10.3390/magnetochemistry9100220.
[43] J.M. Silveyra, J.M.C. Garrido, On the modelling of the anhysteretic magnetization of homogeneous soft magnetic materials, J. Magn. Magn. Mater. 540 (2021) 168430, https://doi.org/10.1016/j.jmmm.2021.168430.
[44] M.P. Weiss, Sur la nature du champ moléculaire, Ann. Phys. (1914) 134–162, https://doi.org/10.1051/anphys/191409010134.

[45] J.M. Silveyra, M.I. González, T.F. González, J.M.C. Garrido, MagAnalyst: a MATLAB toolbox foranhysteretic magnetization analysis, IEEE Trans. Magn. (2024), https://doi.org/10.1109/TMAG.2024.3408681.
[46] A. Wederni, J. Daza, W. Ben Mbarek, J. Saurina, L. Escoda, J.-J. Suñol, Crystal structure and properties of Heusler alloys: a comprehensive review, Metals 14 (2024) 688, https://doi.org/10.3390/met14060688.
[47] P. Lázpita, M. Sasmaz, E. Cesari, J. Barandiarán, J. Gutiérrez, V. Chernenko, Martensitic transformation and magnetic field induced effects in Ni42Co8Mn39Sn11 metamagnetic shape memory alloy, Acta Mater. 109 (2016) 170–176, https://doi.org/10.1016/j.actamat.2016.02.046.
[48] K.M. Krishnan, Fundamentals and applications of magnetic materials, Oxford University Press, 2016, https://doi.org/10.1093/acprof:oso/9780199570447.003.0006.
[49] A. Deltell, L. Escoda, J. Saurina, J.J. Suñol, Martensitic transformation in Ni-Mn-Sn-Co Heusler alloys, Metals 5 (2015) 695–705, https://doi.org/10.3390/met5020695.